(a) PG Ψ, shelf $s = 2$

(b) layer model, steering along f/H

$\times 10^{-7}$

1
0.5
0
−0.5
−1

1.0
0.5
0.0
−0.5
−1.0
y

−2
−1
0
1
2
x

(c) section across the plume, $x = -0.5$

$\Psi / \max|\Psi_{\mathrm{PG}}|$

shelf $s = 1$
shelf $s = 2$
shelf $s = 5$
layer model
shelf

0.5
0.0
−0.5
−1.0

−1.0
−0.5
0.0
0.5
1.0
y

(d) PG against the layer model

relative L_2 difference in Ψ

10^0
10^{-2}

deep, f/H
deep, $f/(H - \delta_E/2)$
shelf, f/H
shelf, $f/(H - \delta_E/2)$

1
2
5
shelf $s = H_{\mathrm{shelf}}/\delta_E$